



YEAR 11 MATHEMATICS EXTENSION 1

Task 2

12 June 2025

General instructions

- Working time – 50 min.
- Write using a blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NSB approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **All 4** questions.

Class (please tick)

- ☐ 11 MX1.A3 – Mr Umakanthan
- ☐ 11 MX1.B7 – Mr Ireland
- ☐ 11 MX1.C7 – Ms Moss
- ☐ 11 MX1.D5 – Dr Vranešević
- ☐ 11 MX1.E6 – Mr Berry
- ☐ 11 MX1.F4– Ms Lee
- ☐ 11 MX1.G7 – Mr Lott

STUDENT NUMBER

Marker's use only.

QUESTION	MCQ	1	2	3	4	Total	%
MARKS	$\overline{5}$	$\overline{7}$	$\overline{15}$	$\overline{6}$	$\overline{10}$	$\overline{43}$	$\overline{100}$

Use the multiple questions answer sheet provided to answer Q1 through to Q5

1. There are 4 letters and 4 addressed envelopes. If each letter is randomly placed in an envelope, then the number of ways in which only one letter goes in the wrong envelope is

A. 43

B. $\binom{4}{1} + \binom{4}{2} + \binom{4}{3} \binom{4}{4}$

C. $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3}$

D. 0

2. $(2x + 1)$ and $(x - 2)$ are the factors of $(2x^3 + px^2 + q)$

What is the value of $(2p + q)$?

A. -10

B. $-\frac{22}{3}$

C. 10

D. $\frac{22}{3}$

3. The quadratic expression $x^2 - 14x + 9$ factorises as $(x - \alpha)(x - \beta)$, where α and β are positive real numbers.

Which quadratic expression can be factorised as $(x - \sqrt{\alpha})(x - \sqrt{\beta})$?

A. $x^2 + \sqrt{10}x + 3$

B. $x^2 - \sqrt{20}x + 3$

C. $x^2 - \sqrt{17}x + 81$

D. $x^2 + \sqrt{19x + 81}$

4. The non-zero constant k is chosen so that the coefficient of x^6 in the expansion of $(1 + kx^2)^7$ and $(k + x)^{10}$ are equal. What is the value of k ?

A. $\frac{1}{6}$

B. 6

C. $\frac{\sqrt{6}}{6}$

D. $\sqrt{6}$

5. The polynomial $n^2x^{2n+3} - 25nx^{n+1} + 150x^7$ has $x^2 - 1$ as a factor

A. for no values of n

B. for $n=10$ only

C. for $n=15$ only

D. for $n = 10$ and $n=15$ only

Question 6. (7 Marks) Use the Question 1 Writing Booklet

- (a) Find the coefficient of x^{10} in the expansion of $(3 + 2x)^{13}$ 1
- (b) Find the independent term of x in the expansion of $\left(3x^2 - \frac{1}{x}\right)^{18}$ 2
- (c) Find the coefficient of x^8 in the expansion of $(x^2 - 2x - 3)^5$. 2
- (d) Find the values of n and r if $\binom{n+1}{7} = \binom{12}{5} + \binom{12}{r}$ (No working required). 2.

Question 7. (15 Marks) Use the Question 2 Writing Booklet

- (a) The number of selections of r objects from an unknown number of different objects is 56 and the total number of different arrangements of the r objects from the same unknown number of different objects is 6720. Find the value of r . 2
- (b) A school library has 3 books by Dickens, 4 books by Shakespeare and 2 books by Montgomery. Find the number of ways of arranging the books on a shelf if
- (i) The books written by the same author are indistinguishable. 1
- (ii) The books written by the same author are distinguishable and must be together. 2
- (iii) The books written by the same author are distinguishable and all the 3 books by Dickens must be separated. 2
- (c) Find how many ways different arrangements can be made using all the letters of the word **CHOCOLATE**, if
- (i) the vowels must be together. 2
- (ii) the 2 letters **O** must be separated. 2
- (d) What is the sum of all 4 digit numbers formed from 3, 5, 5, 6, using the digits only once? 2
- (e) How many 4 digit numbers can be formed with at least one digit repeated? 2

Question 8. (6 Marks) Use the Question 3 Writing Booklet

- (a) Consider the expansion of $\left(\frac{2x}{3} + \frac{3}{2x}\right)^n$ in descending power of x . 3

If the ratio of the coefficient of the 4th term to that of the 5th term is 16 : 45, calculate the value of n .

- (b) Use the expansion of $\left(2 \sin x + \frac{1}{3 \cos x}\right)^{10}$ for $0 \leq x < \frac{\pi}{2}$, to show that 3

$$\binom{10}{10}3^{10} + \binom{10}{9}3^9 + \binom{10}{8}3^8 + \dots + \binom{10}{1}3 + \binom{10}{0} = 2^{20}$$

Question 9. (10 Marks) Use the Question 4 Writing Booklet

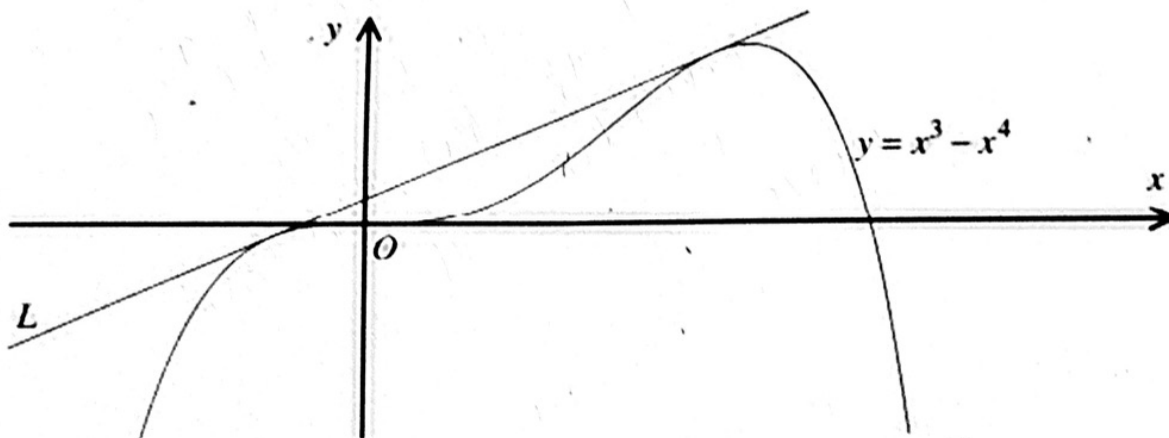
- (a) The polynomial $P(x) = x^3 - 6x^2 + ax + b$ has a root of multiplicity 3. Find the values of a and b . 3

- (b) α and β are the zeros of $3x^2 - 5x + 7 = 0$ 3

Find the value of $3(\alpha^5 + \beta^5) - 5(\alpha^4 + \beta^4) + 7(\alpha^3 + \beta^3)$

- (c) The figure below shows the curve C with equation $y = x^3 - x^4$. 4

The straight line L is a tangent to the C, at two distinct points. Determine the equation of L.



End of Paper

Suggested solution,

∴ Answer: ☐ (D)

Matching the coefficients,

pc - 522 + 37

$$2 = 2a, \quad 3a - 2 = 0$$

$$\Rightarrow a = \frac{2}{3}, p = -\frac{13}{3}, q = \frac{4}{3}$$

$$\Rightarrow 2p + q = -\frac{26}{3} + \frac{4}{3} = -\frac{22}{3}$$

B

$$p(-\frac{1}{2}) = 0$$

$$\Rightarrow 2(-\frac{1}{2})^3 + p(-\frac{1}{2}) + q = 0$$

$$\rightarrow p + 4q = 1 \quad \text{--- (1)}$$

$$p(2) = 0$$

$$\Rightarrow 2(2)^3 + p(2)^2 + q = 0$$

$$\Rightarrow 4pt_2 = -16 \quad \text{--- (2)}$$

solving p and q
simultaneously,

$$p = \frac{13}{3}, q = \frac{4}{3}$$

$$\Rightarrow 2p + q = -\frac{26}{3} + 4$$

3) $x^2 - 14x + 9 = (x - \alpha)(x - \beta)$
 $\alpha + \beta = 14, \alpha\beta = 9$

$$\Rightarrow \alpha + \beta = 14, \alpha\beta = 9$$

$$\Rightarrow (\sqrt{\alpha} + \sqrt{\beta})^2 \sqrt{\alpha\beta} = \alpha + \beta + 2\sqrt{\alpha\beta} = 14 + 6 = 20$$

$$\Rightarrow \frac{\sqrt{a} + \sqrt{b}}{\sqrt{ab}} = \frac{\sqrt{2a}}{\sqrt{9}} = 3$$

$\therefore$ The quadratic eqⁿ is

$$x^2 - \sqrt{20}x + 3 = 0 \quad \textcircled{B}$$

4) The coeff. of x^6 in the expansion of $(1+ax^2)^7$ is $\binom{7}{3}k^3$.
The coeff. of x^{10} in the expansion of $(x+a)^{10}$ is $\binom{10}{6}k^4$.

$$\Rightarrow \frac{\binom{7}{3} k^3}{\binom{10}{6} k^4} = 1 \Rightarrow \frac{7!}{3!4!} \cdot \frac{6!4!}{10!k^4} = 1$$

$$\Rightarrow k^3 = 6k^4 \Rightarrow k^3(6k-1) = 0$$

$$K \neq 0 \Rightarrow K = \frac{1}{6} \quad \textcircled{A}$$

5) $P(x) = n^2 x^{2n+3} - 25n x^{n+1} + 150 x^7$ (x^2-1) is a factor

$$\Rightarrow P(1) = 0 \Rightarrow n^2 - 25n + 150 = 0$$

and

$$\Rightarrow (n-10)(n-15) = 0$$

$$\Rightarrow n \leq 10, n \geq 15$$

$$\Rightarrow P(-1) = 0 \Rightarrow -n^2 + 25n - 150 = 0$$

$$\Rightarrow n^2 - 25n + 150 = 0$$

Test $n=10$

$$100 - 250 + 150 = 0 \text{ True.}$$

Test $n=15$

$$225 - 25 \times 15 + 150 \neq 0 \text{ False}$$

$$\therefore n=10 \text{ only } \boxed{B}$$

~~The coefficient of the~~
6) $(r+1)$ th term in the expansion of $(3+2x)^{13}$ is

$$T_{r+1} = \binom{13}{r} 3^{13-r} 2^r x^r$$

for the coeff. of x^{10} , $r=10$

$$\therefore \text{The required coeff} = \binom{13}{10} 3^3 \cdot 2^{10} =$$

.1 mark for correct term and coefficient

b) The $(r+1)$ th term in the expansion of $(3x^2 - \frac{1}{x})^{18}$ is

$$T_{r+1} = \binom{18}{r} (3x^2)^{18-r} \left(-\frac{1}{x}\right)^r$$

$$= \binom{18}{r} 3^{18-r} \cdot (-1)^r \cdot x^{36-3r}$$

for independent term $36-3r=0$

$$\Rightarrow r=12$$

$$\therefore \text{The required term is } \binom{18}{12} 3^6 \cdot (-1)^{12}$$

$$= \binom{18}{12} 3^6$$

.1 mark for correct coeff
.1 mark for correct value for r

$$6) (c) (x^2 - 2x + 3)^5$$

$$= [x^2 - (2x + 3)]^5$$

$$\text{The } (r+1)\text{th term } T_{r+1} = \binom{5}{r} (x^2)^{5-r} (-1)^r (2x+3)^r$$

$$= \binom{5}{r} (-1)^r x^{10-2r} \cdot \binom{r}{p} (2x)^{r-p} (3)^p$$

$$= \binom{5}{r} \binom{r}{p} (-1)^r 2^{r-p} 3^p (x)^{10-r+p}$$

$$\text{For } x^8$$

$$10 - r + p = 8$$

$$\Rightarrow r + p = 2$$

$$\Rightarrow \left. \begin{matrix} r=2 \\ p=0 \end{matrix} \right\} \left. \begin{matrix} r=1 \\ p=1 \end{matrix} \right\}$$

$\Rightarrow$ required answer is

$$\binom{5}{2} \binom{2}{0} (-1)^2 2^2 3^0 + \binom{5}{1} \binom{2}{1} (-1)^1 2^1 3^1$$

$$= 40 - 15$$

$$= \underline{\underline{25}}$$

$$(d) \quad n=12, r=6 \quad (\text{note: } \binom{12}{5} = \binom{12}{7})$$

$$\left[{}^{n+1}C_r = {}^nC_r + {}^nC_{r-1} \right]$$

- 1 mark for correct values of n and r
- 1 mark for correct value of $\binom{12}{5}$

7 (i) (1) DISABLE has 7 distinct letters
 $\therefore$ no. of ways to choose 5 letters $= {}^7C_5 = 21$

• 1 mark for correct answer

(ii) no. of ways to arrange 7 distinct letters
 $= 7P_5 = 2520$

• 1 mark for correct answer

(b) (i) arrange 3 groups Dickens (3), Shakespeare (4), Montagu (2)
 $\therefore$ no. of ways to arrange the groups

$$= \frac{9!}{3!4!2!} = \underline{\underline{1260}}$$

• 1 mark for correct answer

(ii) Total books = 9

First arrange the 6 non-Dickens books $= 6! = 720$
 Now there are 7 spaces between/around these books
 to place the 3 Dickens books (so that no two are
 together) $= {}^7C_3 = 35$

Arrange the 3 Dickens books $= 3! = 6$

$$\therefore \text{Total} = 720 \times 35 \times 6 = \underline{\underline{151200}}$$

• 1 mark for correct method
 • 1 mark for correct calculation (groups + arrangements)

7 a (ii) Choosing 3 consonants from 4 $= \binom{4}{3}$

" 2 vowels " 3 $= \binom{3}{2}$

" 3 consonants and 2 vowels $= \binom{4}{3} \times \binom{3}{2}$

$\therefore$ Arranging these 5 letters in 5 places $= \binom{4}{3} \binom{3}{2} \times 5!$

$= 960$
 • 1 mark for correct expression for choosing 3 consonants + 2 vowels
 • 1 mark for correct final answer

7(c)(i) CHOCOLATE

vowel: O O A E Consonants: C H C L T

Treat vowels as a block.

5 consonants + 1 vowel block = 6 items

(1 mark: for correct grouping and arrangement)
Arrange these 6 items $\Rightarrow \frac{6!}{2!}$ ways (as there are 2 C's)

Arrange the vowels within the block = $\frac{4!}{2!}$

(1 mark: for correct calculation) $\therefore$ Total $\frac{6!}{2!} \times \frac{4!}{2!} = \underline{4320}$

c(ii) arrangement without restriction = $\frac{9!}{2! \cdot 2!}$

" with O's together = $\frac{8!}{2!}$

$\therefore$ So arrangements with O's separated =

(1 mark: correct method (total - together))
(1 mark: correct calculation)
$$\frac{9!}{2! \cdot 2!} - \frac{8!}{2!} = \underline{70560}$$

7(d) Digits 3, 5, 5, 6 (5 is repeated)

no. of unique numbers = $\frac{4!}{2!} = 12$

Sum of digits = $3 + 5 + 5 + 6 = 19$

Each digit appears in each place (thousands, hundreds, tens, units) = $\frac{12}{4} = 3$ times

Sum: $3 \times (1000 + 100 + 10 + 1) \times 19 = \underline{63327}$

(1 mark: correct method (digit sums, positions))

(1 mark: correct calculation)

7(e)

digits (0-9)

Total 4 digit numbers = $9 \times 10 \times 10 \times 10 = 9000$
(from 1000 to 9999)

Numbers with all digits different

- First digit 9 choices
- Second digit 9 (excluding first digit)
- Third 8
- Fourth 7

Total: $9 \times 9 \times 8 \times 7 = 4536$

So, numbers with at least one digit repeated:
 $9000 - 4536$
 $= 4464$

~~(or simply)~~

Q 8 (a) In the descending powers of the expansion of $\left[\frac{x}{3} + \frac{3}{x}\right]^n$

$$\text{The 4th term} = T_4 = \binom{n}{3} \left(\frac{x}{3}\right)^{n-3} \left(\frac{3}{x}\right)^3$$

$$= \binom{n}{3} \frac{2^{n-6}}{3^{n-6}}$$

$$\text{The 5th term} = T_5 = \binom{n}{4} \left(\frac{x}{3}\right)^{n-4} \left(\frac{3}{x}\right)^4$$

$$= \binom{n}{4} \frac{2^{n-8}}{3^{n-8}}$$

$$T_4 : T_5 = 16 : 45$$

$$\Rightarrow \frac{\binom{n}{3} \frac{2^{n-6}}{3^{n-6}}}{\binom{n}{4} \frac{2^{n-8}}{3^{n-8}}} = \frac{16}{45}$$

$$\Rightarrow \frac{16}{9(n-3)} = \frac{16}{45}$$

$$\Rightarrow n-3 = 5$$

$$\Rightarrow \underline{\underline{n=8}}$$

- 1 mark for correct generation and ratio setup
- 1 mark for correct simplification
- 1 mark for correct value of n

8(b) In the expansion of $(2\sin x + \frac{1}{3\cos x})^{10}$, the $(r+1)^{\text{th}}$ term

$$T_{r+1} = \binom{10}{r} (2\sin x)^{10-r} \left(\frac{1}{3\cos x}\right)^r$$

when $x = 45^\circ$

$$T_{r+1} = \binom{10}{r} \frac{2^{10-r}}{3^r} \cdot \frac{(\sqrt{2})^r}{(\sqrt{2})^{10-r}}$$

$$= \binom{10}{r} \frac{2^{10-r}}{3^r} \cdot \frac{1}{(\sqrt{2})^{10-2r}}$$

$$= \binom{10}{r} \frac{2^{10-r}}{3^r} \cdot \frac{1}{(2)^{5-r}}$$

$$= \binom{10}{r} \frac{2^5}{3^r}$$

when $x = 45^\circ$

$$\left(2\sin x + \frac{1}{3\cos x}\right)^{10} = \left(\sqrt{2} + \frac{\sqrt{2}}{3}\right)^{10} = \left(\frac{4\sqrt{2}}{3}\right)^{10} = \frac{2^{25}}{3^{10}}$$

$$\Rightarrow \frac{2^{25}}{3^{10}} = \binom{10}{0} 2^5 + \binom{10}{1} \frac{2^5}{3} + \binom{10}{2} \frac{2^5}{3^2} + \dots + \binom{10}{10} \frac{2^5}{3^{10}}$$

$$\Rightarrow 2^{20} = \binom{10}{0} 3^{10} + \binom{10}{1} 3^9 + \binom{10}{2} 3^8 + \dots + \binom{10}{10}$$

$$\Rightarrow 2^{20} = \binom{10}{0} 3^{10} + \binom{10}{1} 3^9 + \dots + \binom{10}{10}$$

- 1 mark for correct expansion
- 1 mark for correct substitution
- 1 mark for correct conclusion

Q9 a) $P(x) = x^3 - 6x^2 + ax + b$

$P(x)$ has a root of multiplicity 3 at (say) $x = \alpha$

$\Rightarrow P(\alpha) = 0, P'(\alpha) = 0$ and $P''(\alpha) = 0$

$P(\alpha) = 0 \Rightarrow \alpha^3 - 6\alpha^2 + a\alpha + b = 0$ — (1)

$P'(\alpha) = 0 \Rightarrow 3\alpha^2 - 12\alpha + a = 0$ — (2)

$P''(\alpha) = 0 \Rightarrow 6\alpha - 12 = 0$ — (3)

From (3) $\alpha = 2$

Sub $\alpha = 2$ into (2) gives $a = 12$

Sub $\alpha = 2$ and $a = 12$ into (1) gives $b = -8$

- 1 mark for identification of $P(x) = P'(x) = P''(x) = 0$
- 1 mark for correct values of a
- 1 mark for correct value of b

Alternative method

$P(x) = x^3 - 6x^2 + ax + b \equiv (x - \alpha)^3$

matching the coefficients,

$-3\alpha = 6 \Rightarrow \alpha = 2$

$3\alpha^2 = a \Rightarrow a = 12$

$-\alpha^3 = b \Rightarrow b = -8$

- 1 mark for correct identification of this form
- 1 mark for correct value for a
- 1 mark for correct value for b

Q9 (b)

$$3x^2 - 2(a+b+c)x + (ab+bc+ca) = 0$$

Method 1 $\Delta = 0$

Method 2 $P(x) = P'(x) = 0$

both will give

$$(a+b+c)^2 - 3(ab+bc+ca) = 0$$

$$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = 0$$

$$\Rightarrow \frac{1}{2} [(a-b)^2 + (b-c)^2 + (c-a)^2] = 0$$

as sum of 3 squares = 0

each must be equal to zero

$$\Rightarrow a = b = c$$

- 1 mark for set up: $\Delta = 0$ or $P(x) = P'(x) = 0$
- 1 mark for simplification to sum of 3 squares
- 1 mark for correct conclusion

Q9 (c) Let $L: y = mx + b$

$\Rightarrow x^3 + mx + b = 0$ has roots $\alpha, \alpha, \beta, \beta$

L is touching the curve at 2 different points

$\therefore$ 2 coincident roots

$$\Rightarrow \text{sum of roots} = -1 \Rightarrow 2(\alpha + \beta) = -1 \Rightarrow \alpha + \beta = -\frac{1}{2} \quad (1)$$

$$\Rightarrow \text{sum of roots taken 2 at a time} = 0$$

$$\Rightarrow \alpha^2 + \alpha\beta + \beta^2 = 0 \quad (2)$$

$$\Rightarrow \text{sum of roots taken 3 at a time} = -m$$

$$\Rightarrow 2(\alpha^2\beta + \alpha\beta^2) = -m \quad (3)$$

$$\Rightarrow \text{product of roots} = b$$

$$\Rightarrow \alpha^2\beta^2 = b \quad (4)$$

Solving equations (1), (2), (3) and (4) simultaneously gives

$$m = \frac{1}{8} \text{ and } b = \frac{1}{64}$$

$$\therefore L: y = \frac{1}{8}x + \frac{1}{64}$$

- 1 mark for identifying that there are 2 coincident roots
- 1 mark for writing the 4 relationships correctly between roots and coefficients
- 1 mark for correct values of m and b
- 1 mark for conclusion for L